- Compute Match — Product Overview
  - What is Compute Match?
  - Who uses Compute Match?
  - The core idea in five steps
  - Key business objects
  - Important timers (defaults)
  - Technology at a glance
  - Where to get help

Compute Match — Product Overview

What is Compute Match?

Compute Match, also called the **Compute MatchMake System** (internal project key **NCPMS**, code name **Clustruffle**), is an NVIDIA web platform that connects **customers** who need GPU compute capacity with **NVIDIA Cloud Providers (NCPs)** who own and operate GPU infrastructure.

In one sentence: customers find GPU clusters, send reservation requests, and NCPs accept, negotiate, and contract those requests, while NVIDIA Platform Admins oversee the marketplace.

Who uses Compute Match?

There are three personas:

Persona	Who they are	What they do
Customer (also called **Offtaker** or requester)	An organization that needs GPU compute	Browse available GPU clusters, submit reservation requests (to one or several NCPs), submit custom requests, track and cancel requests

Persona	Who they are	What they do
NCP (NVIDIA Cloud Provider / GPU Compute Provider)	An organization that operates GPU data centers	Upload GPU capacity via CSV, review incoming reservation requests, accept, reject, counteroffer, and confirm contracts
Platform Admin (NVIDIA operations / business admins)	NVIDIA staff running the marketplace	Explore global capacity, review future capacity uploads, onboard NCPs, approve NCP access requests, broker custom requests, extend deadlines, view analytics dashboards

A single user can hold both customer and NCP access; they switch between them with **Switch profile to** in the account menu.

The core idea in five steps

1. **NCPs publish capacity.** An NCP uploads a CSV describing how many GPUs of which platform (for example H100, B200, GB200) are available, in which country and city, and from which date.
2. **Customers search.** A customer opens **Find a Cluster to Reserve**, filters by platform, region, country, GPU count, tenure, and earliest start date, and sees matching clusters on a world map and as cards.
3. **Customers request.** The customer submits a **reservation request** for one cluster, or for up to 3 clusters from different NCPs at once (multi-NCP request). If nothing fits, they submit a **custom request** that NVIDIA admins broker manually.
4. **NCPs respond.** Each selected NCP receives an email with a secure **magic link**. The NCP accepts (moving the request to **In Process**), rejects, or counteroffers, and finally clicks **Confirm Contract**.
5. **Contract.** The first NCP to contract wins; the other NCPs' offerings on the same request are cancelled as superseded. Capacity is allocated, and the deal is synced to Salesforce as an Offtake record.

Key business objects

Object	Meaning
GPU Offering / GPU Capacity	A cluster an NCP makes available: GPU platform, hosting provider, region, country, city, GPU count and availability date windows
Capacity type	`AVAILABLE` (current capacity, starts within 90 days of upload, published right away) or `FUTURE` (starts more than 90 days out, needs Platform Admin review)
Reservation Request	A customer's request to reserve GPU capacity
Reservation Request Offering	One NCP's slice of a reservation request; a multi-NCP request has one per NCP, each with its own status and deadlines
Proposal	One version of the reservation terms (dates, GPU count, platform, region, price). The customer's original request is the first proposal; NCP counteroffers add more
Custom Request	A request without a specific cluster; NVIDIA admins move it through Open, In Review, Matched or Unserviceable
Allocation	Capacity consumed once a request is contracted
Import Job	One CSV capacity upload and its processing status
Access Request	A pending request from a partner (for example via the NPN portal) to join as an NCP

Important timers (defaults)

- An NCP has **7 days** to accept a request routed to them (acceptance window).
- After accepting, the NCP and customer have **45 days** to reach a contract (contract negotiation window).
- Reminder emails are sent **3 days** before each deadline.
- A magic link is valid for **1 hour**, and can be regenerated for up to **7 days** after routing.
- Custom requests never expire automatically.
- Platform Admins can extend either window by **1 to 200 days**.

Technology at a glance

- **Frontend:** Next.js 15, React 19, Ant Design 6, Tailwind CSS 4, NextAuth 5 with Keycloak.
- **Backend:** Python 3.13, FastAPI, Domain-Driven Design with hexagonal architecture, SQLAlchemy, PostgreSQL, Alembic migrations.
- **Workflows:** Temporal (Temporal Cloud in shared environments) with a Python workflow worker.
- **Identity:** Keycloak with NVIDIA Starfleet SSO, custom Keycloak extensions.
- **Analytics:** Apache Superset dashboards.
- **Platform:** Google Cloud — GKE Autopilot, Cloud SQL PostgreSQL, Memorystore Redis, Secret Manager; deployed with Terraform and Helm through GitLab CI.
- **Integrations:** Salesforce (Offtake sync and partner-domain sync), email notifications.

Where to get help

- In the app: **Page Help** (the `?` icon) for context help and **User Manual** (book icon) for the full role-filtered manual.
- Support email for access problems: computematchmake@nvidia.com.
- Developer docs: the `docs/` folder in the repository, also published as the Compute Match documentation site.